Interview Presentations

Antti Käenmäki

Tampere, 14th January 2025

Interview presentations

1. **Teaching Demonstration (20 min)**

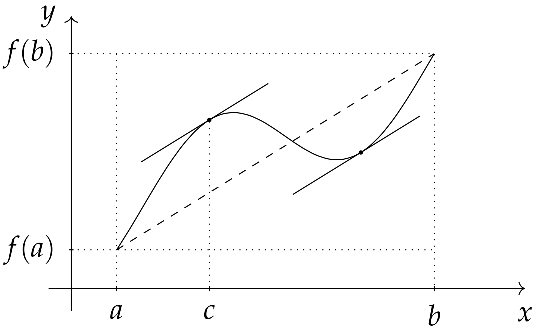
2. **Research Presentation (15 min)**

3. **Questions & Answers (15 min)**

Mean value theorem

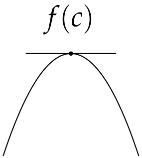
Mean Value Theorem. Let $f\colon [a,b] \to \mathbb{R}$ be a continuous function on the proper closed interval $[a,b]$, and differentiable on the open interval (a,b). Then there exists some $c \in (a,b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Mean value theorem

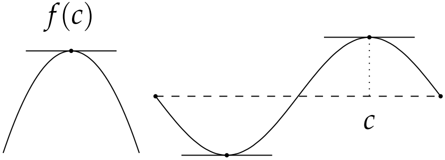
The idea in the proof can be illustrated in three steps:



❶ Prove that the derivative at a maximal point is zero.

Mean value theorem

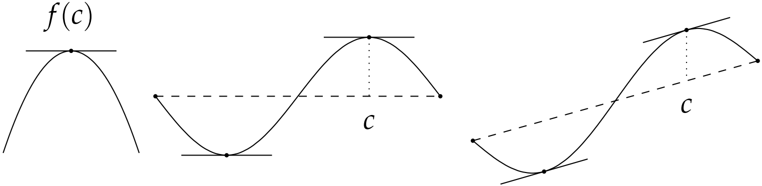
The idea in the proof can be illustrated in three steps:



1. Prove that the derivative at a maximal point is zero.
2. Use (1) to show that if $f(a) = f(b)$, then the derivative at extremal points is zero.

Mean value theorem

The idea in the proof can be illustrated in three steps:



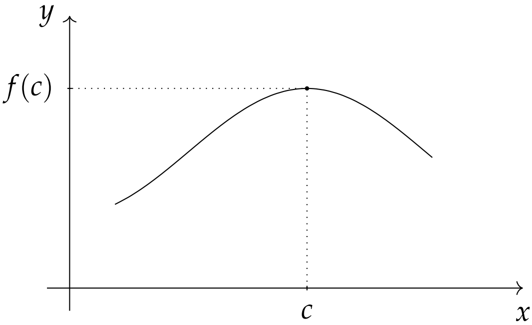
1. Prove that the derivative at a maximal point is zero.
2. Use (1) to show that if $f(a) = f(b)$, then the derivative at extremal points is zero.
3. Use (2) to conclude that an analogous claim holds if the graph is sheared vertically.

Mean value theorem

Lemma. Let $f : (a, b) \to \mathbb{R}$ be differentiable at $c \in (a, b)$. If

$$f(x) \leqslant f(c)$$

for all $x \in (a, b)$, then $f'(c) = 0$.



Mean value theorem

Proof: Notice that, by the assumption, we have $f(x) - f(c) \leqslant 0$ for all $x \in (a, b)$.

Mean value theorem

Proof: Notice that, by the assumption, we have $f(x) - f(c) \leqslant 0$ for all $x \in (a, b)$.

- If $x \in (c, b)$, then $x - c > 0$ and

$$\frac{f(x) - f(c)}{x - c} \leqslant 0.$$

By letting $x \downarrow c$, we see that $f'(c) \leqslant 0$.

Mean value theorem

Proof: Notice that, by the assumption, we have $f(x) - f(c) \leqslant 0$ for all $x \in (a, b)$.

- If $x \in (c, b)$, then $x - c > 0$ and
$$\frac{f(x) - f(c)}{x - c} \leqslant 0.$$
By letting $x \downarrow c$, we see that $f'(c) \leqslant 0$.

- Similarly, if $x \in (a, c)$, then $x - c < 0$ and
$$\frac{f(x) - f(c)}{x - c} \geqslant 0.$$
By letting $x \uparrow c$, we get $f'(c) \geqslant 0$.

Mean value theorem

Proof: Notice that, by the assumption, we have $f(x) - f(c) \leqslant 0$ for all $x \in (a, b)$.

- If $x \in (c, b)$, then $x - c > 0$ and
$$\frac{f(x) - f(c)}{x - c} \leqslant 0.$$
By letting $x \downarrow c$, we see that $f'(c) \leqslant 0$.

- Similarly, if $x \in (a, c)$, then $x - c < 0$ and
$$\frac{f(x) - f(c)}{x - c} \geqslant 0.$$
By letting $x \uparrow c$, we get $f'(c) \geqslant 0$.

We conclude that $f'(c) = 0$. $\square$

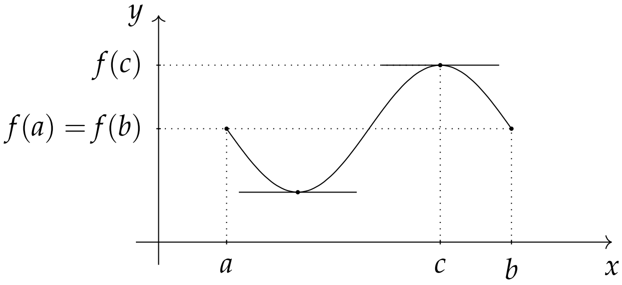
Mean value theorem

Rolle's Theorem. Let $f : [a,b] \to \mathbb{R}$ be a continuous function on the proper closed interval $[a,b]$, and differentiable on the open interval (a,b). If

$$f(a) = f(b),$$

then there exists some $c \in (a,b)$ such that

$$f'(c) = 0.$$



Mean value theorem

Proof: Since f is continuous on $[a, b]$, it attains both its maximum and minimum in $[a, b]$.

Mean value theorem

Proof: Since f is continuous on $[a, b]$, it attains both its maximum and minimum in $[a, b]$.

- If these are both attained at the endpoints of $[a, b]$, then f is constant on $[a, b]$ and $f'(x) = 0$ for all $x \in (a, b)$.

Mean value theorem

Proof: Since f is continuous on $[a, b]$, it attains both its maximum and minimum in $[a, b]$.

- If these are both attained at the endpoints of $[a, b]$, then f is constant on $[a, b]$ and $f'(x) = 0$ for all $x \in (a, b)$.
- Suppose then that the maximum is obtained at $c \in (a, b)$. Since, in particular, f is differentiable at c and

$$f(x) \leqslant f(c)$$

for all $x \in (a, b)$, Lemma implies $f'(c) = 0$.

Mean value theorem

Proof: Since f is continuous on $[a, b]$, it attains both its maximum and minimum in $[a, b]$.

- If these are both attained at the endpoints of $[a, b]$, then f is constant on $[a, b]$ and $f'(x) = 0$ for all $x \in (a, b)$.
- Suppose then that the maximum is obtained at $c \in (a, b)$. Since, in particular, f is differentiable at c and

$$f(x) \leqslant f(c)$$

for all $x \in (a, b)$, Lemma implies $f'(c) = 0$.

- If the minimum is obtained at $c \in (a, b)$, then by considering $-f$, the above argument shows that $f'(c) = 0$. $\square$

Mean value theorem

We use Rolle's Theorem to prove the Mean Value Theorem.

Mean Value Theorem. Let $f\colon [a,b] \to \mathbb{R}$ be a continuous function on the proper closed interval $[a,b]$, and differentiable on the open interval (a,b). Then there exists some $c \in (a,b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Mean value theorem

We use Rolle's Theorem to prove the Mean Value Theorem.

Mean Value Theorem. Let $f \colon [a, b] \to \mathbb{R}$ be a continuous function on the proper closed interval $[a, b]$, and differentiable on the open interval (a, b). Then there exists some $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof: Let $s \in \mathbb{R}$ and define $g \colon [a, b] \to \mathbb{R}$ by setting

$$g(x) = f(x) - sx$$

for all $x \in [a, b]$. Since f is continuous on $[a, b]$ and differentiable on (a, b), the same is true for g and $g'(x) = f'(x) - s$.

Mean value theorem

We use Rolle's Theorem to prove the Mean Value Theorem.

Mean Value Theorem. Let $f\colon [a,b] \to \mathbb{R}$ be a continuous function on the proper closed interval $[a,b]$, and differentiable on the open interval (a,b). Then there exists some $c \in (a,b)$ such that

$$f'(c) = \frac{f(b)-f(a)}{b-a}.$$

Proof: Let $s \in \mathbb{R}$ and define $g\colon [a,b] \to \mathbb{R}$ by setting

$$g(x) = f(x) - sx$$

for all $x \in [a,b]$. Since f is continuous on $[a,b]$ and differentiable on (a,b), the same is true for g and $g'(x) = f'(x) - s$.

We now want to choose s so that $g(a) = g(b)$, in which case g satisfies the assumptions of Rolle's Theorem.

Mean value theorem

A simple calculation shows that

$$\begin{aligned} g(a) = g(b) \quad &\Leftrightarrow \quad f(a) - sa = f(b) - sb \\ &\Leftrightarrow \quad s(b-a) = f(b) - f(a) \\ &\Leftrightarrow \quad s = \frac{f(b) - f(a)}{b-a}. \end{aligned}$$

Mean value theorem

A simple calculation shows that

$$\begin{aligned} g(a) = g(b) \quad &\Leftrightarrow \quad f(a) - sa = f(b) - sb \\ &\Leftrightarrow \quad s(b-a) = f(b) - f(a) \\ &\Leftrightarrow \quad s = \frac{f(b) - f(a)}{b-a}. \end{aligned}$$

With this choice of s, we may apply Rolle's Theorem to g. Doing so, we find $c \in (a, b)$ such that

$$0 = g'(c) = f'(c) - s.$$

Mean value theorem

A simple calculation shows that

$$\begin{aligned} g(a) = g(b) \quad &\Leftrightarrow \quad f(a) - sa = f(b) - sb \\ &\Leftrightarrow \quad s(b-a) = f(b) - f(a) \\ &\Leftrightarrow \quad s = \frac{f(b) - f(a)}{b-a}. \end{aligned}$$

With this choice of s, we may apply Rolle's Theorem to g. Doing so, we find $c \in (a, b)$ such that

$$0 = g'(c) = f'(c) - s.$$

We have thus shown the existence of $c \in (a, b)$ for which

$$f'(c) = s = \frac{f(b) - f(a)}{b-a}$$

as claimed. $\square$

Mean value theorem

The Mean Value Theorem has several applications:

Corollary. Let $f\colon [a,b] \to \mathbb{R}$ be a continuous function on the proper closed interval $[a,b]$, and differentiable on the open interval (a,b).

1. If $f'(x) \geqslant 0$ for all $x \in (a,b)$, then f is increasing on $[a,b]$.
2. If $f'(x) \leqslant 0$ for all $x \in (a,b)$, then f is decreasing on $[a,b]$.
3. If $f'(x) = 0$ for all $x \in (a,b)$, then f is constant.

Mean value theorem

The Mean Value Theorem has several applications:

Corollary. Let $f\colon [a,b] \to \mathbb{R}$ be a continuous function on the proper closed interval $[a,b]$, and differentiable on the open interval (a,b).

1. If $f'(x) \geqslant 0$ for all $x \in (a,b)$, then f is increasing on $[a,b]$.
2. If $f'(x) \leqslant 0$ for all $x \in (a,b)$, then f is decreasing on $[a,b]$.
3. If $f'(x) = 0$ for all $x \in (a,b)$, then f is constant.

Proof: Let $x, y \in [a,b]$ be such that $x < y$. By the Mean Value Theorem, there is $c \in (a,b)$ such that

$$f(y) - f(x) = f'(c)(y-x).$$

In particular, $f'(c)$ and $f(y)-f(x)$ have the same sign. The claims follow. $\square$

Mean value theorem

Example. How well does 1.4 approximate $\sqrt{2}$?

Let $f(x) = \sqrt{x}$ in which case

$$f'(x) = \frac{1}{2\sqrt{x}}$$

for all $x > 0$. Notice that f' is decreasing. Since $1.4^2 = 1.96 < 2$, we have $1.4 < \sqrt{2}$. By the Mean Value Theorem, there is $c \in (1.96, 2)$ such that

$$\begin{aligned}\sqrt{2} - 1.4 &= f(2) - f(1.96) = f'(c)(2 - 1.96) \\ &\leqslant f'(1.96)(2 - 1.96) = \frac{0.04}{2 \cdot 1.4} = \frac{1}{70} \approx 0.014286.\end{aligned}$$

Interview presentations

1. Teaching Demonstration (20 min)

2. Research Presentation (15 min)

3. Questions & Answers (15 min)

Research presentation

- The general theme in my current research is to apply dynamical methods in the study of analytical problems.

Research presentation

- The general theme in my current research is to apply dynamical methods in the study of analytical problems.
- Mathematical control theory concentrates on influencing the behaviour of dynamical models $\rightsquigarrow$ *TAU*.

Research presentation

- The general theme in my current research is to apply dynamical methods in the study of analytical problems.
- Mathematical control theory concentrates on influencing the behaviour of dynamical models $\rightsquigarrow$ *TAU*.
- My research primarily focuses on ergodic theory, fractal geometry, and geometric measure theory.

Research presentation

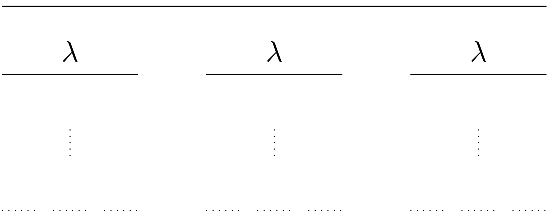
- The general theme in my current research is to apply dynamical methods in the study of analytical problems.
- Mathematical control theory concentrates on influencing the behaviour of dynamical models $\rightsquigarrow$ *TAU*.
- My research primarily focuses on ergodic theory, fractal geometry, and geometric measure theory.
- I am currently focusing on state-of-the-art problems in fractal geometry and some of the methods require understanding on number theory and algebra $\rightsquigarrow$ *TAU*.

Research presentation

- The general theme in my current research is to apply dynamical methods in the study of analytical problems.
- Mathematical control theory concentrates on influencing the behaviour of dynamical models $\rightsquigarrow$ *TAU*.
- My research primarily focuses on ergodic theory, fractal geometry, and geometric measure theory.
- I am currently focusing on state-of-the-art problems in fractal geometry and some of the methods require understanding on number theory and algebra $\rightsquigarrow$ *TAU*.
- I will next present some of the research problems to illustrate my research vision.

Dimension drop conjecture

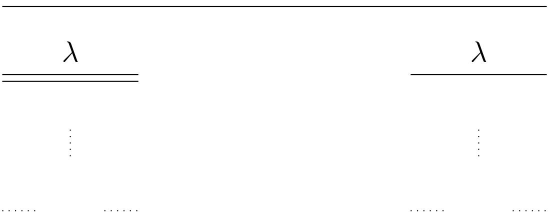
- Consider a self-similar set X which is *strongly separated*:



- It is well-known that the dimension *equals* the similarity dimension: $\dim_{\mathrm{H}}(X) = \log 3 / \log \lambda^{-1} =: \dim_{\mathrm{sim}}(X)$.

Dimension drop conjecture

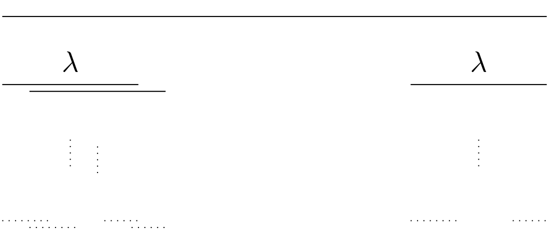
- Let us then assume that the construction has *exact overlaps*:



- In this case the dimension *drops* below the similarily dimension: $\dim_{\mathrm{H}}(X) = \log 2/\log\lambda^{-1} < \dim_{\mathrm{sim}}(X)$.

Dimension drop conjecture

- Let the construction be *overlapping* but not exactly:



- In this situation, there are no known examples with $\dim_{\mathrm{H}}(X) < \dim_{\mathrm{sim}}(X)$.

- The *dimension drop conjecture* claims that the exact overlapping is the only mechanism for the dimension to drop below the similarity dimension.

Dimension drop conjecture

- Hochman (2014) has shown that the dimension drop conjecture holds in the class of the self-similar sets satisfying the *exponential separation*.

Dimension drop conjecture

- Hochman (2014) has shown that the dimension drop conjecture holds in the class of the self-similar sets satisfying the *exponential separation* .

- If the self-similar set is defined by *algebraic parameters* , then it is exponentially separated.

Dimension drop conjecture

- Hochman (2014) has shown that the dimension drop conjecture holds in the class of the self-similar sets satisfying the *exponential separation*.
- If the self-similar set is defined by *algebraic parameters*, then it is exponentially separated.
- Hochman conjectured in his ICM talk that his result covers all the cases.

Dimension drop conjecture

- Hochman (2014) has shown that the dimension drop conjecture holds in the class of the self-similar sets satisfying the *exponential separation* .
- If the self-similar set is defined by *algebraic parameters* , then it is exponentially separated.
- Hochman conjectured in his ICM talk that his result covers all the cases.

  Bárány-K (2021) proved that Hochman's result is not enough.

Dimension drop conjecture

- Hochman (2014) has shown that the dimension drop conjecture holds in the class of the self-similar sets satisfying the *exponential separation* .

- If the self-similar set is defined by *algebraic parameters* , then it is exponentially separated.

- Hochman conjectured in his ICM talk that his result covers all the cases.

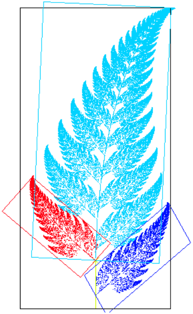
  Bárány-K (2021) proved that Hochman's result is not enough.

- The conjecture is thus open. One can also modify the question:
  - Parametrized families of self-similar sets.
  - Different notions of dimensions.

Self-affine sets

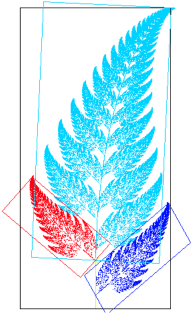
- The self-affine set $X \subset \mathbb{R}^d$ is constructed by using affine maps.

- Hochman-Rapaport (2022) showed that if $X \subset \mathbb{R}^2$ is *strongly irreducible* and satisfies the *exponential separation*, then the dimension is the affinity dimension, $\dim_{\mathrm{H}}(X) = \dim_{\mathrm{aff}}(X)$.



Self-affine sets

- The self-affine set $X \subset \mathbb{R}^d$ is constructed by using affine maps.
- Hochman-Rapaport (2022) showed that if $X \subset \mathbb{R}^2$ is *strongly irreducible* and satisfies the *exponential separation*, then the dimension is the affinity dimension, $\dim_{\mathrm{H}}(X) = \dim_{\mathrm{aff}}(X)$.

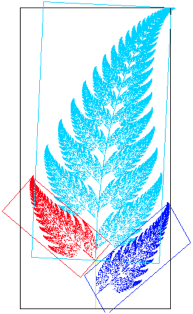


- The proof of the result relies on the existence of an equilibrium state (K 2004), Ledrappier-Young formula (Bárány-K 2017), and Hochman's method.

Self-affine sets

- The self-affine set $X \subset \mathbb{R}^d$ is constructed by using affine maps.

- Hochman-Rapaport (2022) showed that if $X \subset \mathbb{R}^2$ is *strongly irreducible* and satisfies the *exponential separation*, then the dimension is the affinity dimension, $\dim_{\mathrm{H}}(X) = \dim_{\mathrm{aff}}(X)$.



- The proof of the result relies on the existence of an equilibrium state (K 2004), Ledrappier-Young formula (Bárány-K 2017), and Hochman's method.

K-Morris (2025+) found a characterization of equilibrium states in the countably generated case.

- The proof relies on representation theory and algebraic methods.

Self-affine sets

Bárány-K-Yu (2025+) showed that if $X \subset \mathbb{R}^2$ is a dominated self-affine set satisfying separation condition such that $s := \dim_{\mathrm{H}}(X) > 1$, then it cannot be s-Ahlfors regular.

- Can it happen that $\mathcal{H}^s(X) > 0$?

Self-affine sets

Bárány-K-Yu (2025+) showed that if $X \subset \mathbb{R}^2$ is a dominated self-affine set satisfying separation condition such that $s := \dim_{\mathrm{H}}(X) > 1$, then it cannot be s-Ahlfors regular.

- Can it happen that $\mathcal{H}^s(X) > 0$?
- Marstrand's projection theorem says that if $A \subset \mathbb{R}^2$ is a Borel set, then
$$\dim_{\mathrm{H}}(\mathrm{proj}_e(A)) = \min\{1, \dim_{\mathrm{H}}(A)\}$$
for $\mathcal{H}^1$-almost all $e \in S^1$.

Self-affine sets

Bárány-K-Yu (2025+) showed that if $X \subset \mathbb{R}^2$ is a dominated self-affine set satisfying separation condition such that $s := \dim_{\mathrm{H}}(X) > 1$, then it cannot be s-Ahlfors regular.

- Can it happen that $\mathcal{H}^s(X) > 0$?
- Marstrand's projection theorem says that if $A \subset \mathbb{R}^2$ is a Borel set, then
$$\dim_{\mathrm{H}}(\mathrm{proj}_e(A)) = \min\{1, \dim_{\mathrm{H}}(A)\}$$
for $\mathcal{H}^1$-almost all $e \in S^1$.
- For self-affine sets $X \subset \mathbb{R}^2$, is it possible to characterize all directions $e \in S^1$ for which
$$\dim_{\mathrm{H}}(\mathrm{proj}_e(X)) = \min\{1, \dim_{\mathrm{H}}(X)\}?$$

General non-conformal systems

- Instead of affine maps, now we use $C^{1+\alpha}$-maps in the construction.

- Natural idea is to mimick the theory on self-affine
  - Is there a Ledrappier-Young formula in this generality? The associated projections are not anymore orthogonal.
  - What is the structure of equilibrium states?

General non-conformal systems

- Instead of affine maps, now we use $C^{1+\alpha}$-maps in the construction.

- Natural idea is to mimick the theory on self-affine
  - Is there a Ledrappier-Young formula in this generality? The associated projections are not anymore orthogonal.
  - What is the structure of equilibrium states?

Bárány-K (2024) studied the first non-trivial classes of constructions in this regime.

General non-conformal systems

- Instead of affine maps, now we use $C^{1+\alpha}$-maps in the construction.
- Natural idea is to mimick the theory on self-affine
  - Is there a Ledrappier-Young formula in this generality? The associated projections are not anymore orthogonal.
  - What is the structure of equilibrium states?

Bárány-K (2024) studied the first non-trivial classes of constructions in this regime.

- A very recent preprint of Rapaport (2024, Dec 21) extends Hochman's method into non-linear setting in the line.
- Can Rapaport's ideas be extended into higher dimensions?

Interview Presentations

1. Teaching Demonstration (20 min)
2. Research Presentation (15 min)
3. Questions & Answers (15 min)